

## MONCLOVA, Mexico

WMO: 763420

Lat: 26.9003N

Lon: 101.4339W

Elev: 615

StdP: 94.15

Time Zone: -6.00 (MEX)

Period: 95-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 1.2        | 3.4 | -10.4                         | 1.7 | 15.1 | -7.2 | 2.2 | 15.7 | 10.1                  | 22.5 | 8.6 | 16.0 | 2.0                   | 60   | 0.354 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 12.6                   | 40.1            | 23.6 | 38.9 | 23.7 | 37.8 | 23.4 | 28.1                | 36.3 | 26.7 | 35.0 | 25.6 | 34.2 | 3.5                  | 60   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.0                            | 23.0 | 34.6 | 24.4 | 20.9 | 32.8 | 23.1 | 19.3 | 30.3 | 94.8          | 36.1 | 88.0 | 35.2 | 82.5 | 34.2 | 34.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.2           | 7.6  | 6.0 | -1.7                       | 43.2 | 2.6                | 1.7 | -3.6                                               | 44.4 | -5.1       | 45.4 | -6.6       | 46.4 | -8.5       | 47.6 |    |    |
| (5)               |      |     | -2.8                       | 29.4 | 2.6                | 2.7 | -4.7                                               | 31.3 | -6.3       | 32.9 | -7.7       | 34.4 | -9.6       | 36.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 23.7   | 14.6  | 18.3  | 21.1  | 25.4  | 28.5  | 30.5  | 30.3  | 30.5  | 27.6  | 23.8  | 18.8  | 14.5  |  |  |
|                                                                        | DBStd       | 7.04   | 5.49  | 5.00  | 4.82  | 4.16  | 3.42  | 2.33  | 2.24  | 2.07  | 3.11  | 3.82  | 4.41  | 4.76  |  |  |
|                                                                        | HDD10.0     | 47     | 23    | 4     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 3     | 14    |  |  |
|                                                                        | HDD18.3     | 420    | 138   | 59    | 29    | 4     | 0     | 0     | 0     | 0     | 1     | 7     | 46    | 136   |  |  |
|                                                                        | CDD10.0     | 5036   | 166   | 235   | 346   | 462   | 574   | 615   | 628   | 634   | 528   | 428   | 267   | 154   |  |  |
|                                                                        | CDD18.3     | 2368   | 22    | 57    | 115   | 217   | 316   | 365   | 369   | 376   | 278   | 176   | 59    | 17    |  |  |
|                                                                        | CDH23.3     | 29108  | 291   | 628   | 1247  | 2581  | 4033  | 4995  | 4851  | 5022  | 3148  | 1624  | 501   | 189   |  |  |
| (13)                                                                   | CDH26.7     | 15785  | 93    | 259   | 578   | 1391  | 2328  | 2947  | 2781  | 2891  | 1594  | 704   | 167   | 52    |  |  |
| (14)                                                                   | Wind        | WSAvg  | 2.2   | 2.1   | 2.3   | 2.4   | 2.6   | 2.4   | 2.3   | 2.2   | 2.2   | 2.0   | 1.9   | 2.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 367    | 10    | 14    | 13    | 20    | 36    | 59    | 54    | 42    | 81    | 31    | 17    | 14    |  |  |
|                                                                        | PrecMax     | 739    | 40    | 74    | 64    | 88    | 133   | 349   | 288   | 169   | 257   | 79    | 122   | 89    |  |  |
|                                                                        | PrecMin     | 171    | 0     | 0     | 0     | 0     | 0     | 0     | 7     | 1     | 2     | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 135    | 11    | 16    | 17    | 23    | 31    | 76    | 61    | 42    | 72    | 25    | 24    | 18    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 32.0  | 35.1  | 36.7  | 40.2  | 42.1  | 41.8  | 40.2  | 39.9  | 39.1  | 36.8  | 33.9  | 31.2  |  |  |
|                                                                        |             | MCWB   | 18.3  | 20.0  | 18.2  | 21.3  | 23.6  | 23.0  | 24.4  | 23.8  | 23.9  | 22.1  | 20.5  | 19.2  |  |  |
|                                                                        | 2%          | DB     | 29.4  | 32.1  | 34.5  | 38.0  | 39.9  | 39.9  | 38.8  | 38.7  | 36.9  | 34.6  | 30.9  | 28.1  |  |  |
|                                                                        |             | MCWB   | 17.3  | 18.8  | 19.5  | 20.9  | 23.9  | 23.7  | 24.2  | 24.3  | 23.3  | 22.6  | 19.1  | 16.5  |  |  |
|                                                                        | 5%          | DB     | 26.8  | 30.0  | 32.4  | 36.0  | 38.0  | 38.2  | 37.7  | 37.5  | 35.2  | 32.8  | 28.6  | 25.2  |  |  |
|                                                                        |             | MCWB   | 15.9  | 17.8  | 18.7  | 20.6  | 23.4  | 23.5  | 23.7  | 24.0  | 22.9  | 21.9  | 18.9  | 15.9  |  |  |
|                                                                        | 10%         | DB     | 24.0  | 27.3  | 30.1  | 33.8  | 36.1  | 36.8  | 36.2  | 36.3  | 33.8  | 30.8  | 26.2  | 22.9  |  |  |
|                                                                        |             | MCWB   | 14.6  | 16.7  | 18.2  | 20.7  | 22.5  | 23.5  | 23.4  | 23.6  | 22.8  | 21.4  | 17.7  | 15.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 21.5  | 23.7  | 24.7  | 27.6  | 29.2  | 29.8  | 29.1  | 29.7  | 28.0  | 26.4  | 23.1  | 20.6  |  |  |
|                                                                        |             | MCDB   | 29.1  | 33.2  | 33.1  | 35.6  | 38.9  | 37.1  | 36.6  | 36.5  | 34.6  | 33.3  | 30.3  | 29.1  |  |  |
|                                                                        | 2%          | WB     | 18.8  | 20.6  | 21.9  | 24.6  | 26.8  | 26.7  | 27.2  | 27.6  | 26.0  | 24.5  | 20.9  | 18.5  |  |  |
|                                                                        |             | MCDB   | 27.6  | 29.5  | 30.9  | 34.4  | 36.6  | 35.2  | 35.3  | 35.6  | 33.4  | 31.3  | 27.7  | 25.4  |  |  |
|                                                                        | 5%          | WB     | 16.9  | 18.8  | 20.4  | 22.8  | 25.1  | 25.3  | 25.6  | 25.9  | 24.8  | 23.3  | 19.8  | 17.0  |  |  |
|                                                                        |             | MCDB   | 24.6  | 27.5  | 29.2  | 33.0  | 34.9  | 34.5  | 34.2  | 34.5  | 32.2  | 29.5  | 26.1  | 23.3  |  |  |
|                                                                        | 10%         | WB     | 15.1  | 17.4  | 19.3  | 21.4  | 23.9  | 24.3  | 24.5  | 24.7  | 23.8  | 22.4  | 18.6  | 15.7  |  |  |
|                                                                        |             | MCDB   | 22.4  | 25.1  | 27.4  | 31.0  | 32.9  | 33.7  | 33.6  | 33.3  | 31.1  | 28.4  | 25.0  | 21.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.4  | 14.0  | 13.8  | 14.2  | 13.4  | 12.6  | 11.8  | 11.8  | 11.2  | 12.0  | 12.4  | 12.2  |  |  |
|                                                                        |             | MCDBR  | 17.5  | 17.3  | 17.0  | 17.2  | 15.9  | 14.2  | 13.3  | 13.0  | 13.0  | 14.8  | 15.8  | 16.1  |  |  |
|                                                                        |             | MCWBR  | 10.5  | 9.3   | 8.6   | 8.2   | 8.0   | 6.2   | 6.5   | 6.3   | 6.6   | 7.2   | 8.6   | 9.0   |  |  |
|                                                                        | 5% WB       | MCDBR  | 16.6  | 15.8  | 15.4  | 15.2  | 14.4  | 12.8  | 12.3  | 12.2  | 11.6  | 11.8  | 13.9  | 14.9  |  |  |
|                                                                        |             | MCWBR  | 11.5  | 10.0  | 9.8   | 9.5   | 8.9   | 7.3   | 8.0   | 7.9   | 7.1   | 7.1   | 8.5   | 9.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.298 | 0.314 | 0.326 | 0.371 | 0.409 | 0.407 | 0.404 | 0.388 | 0.396 | 0.347 | 0.326 | 0.305 |  |  |
|                                                                        | taud        |        | 2.542 | 2.478 | 2.455 | 2.333 | 2.273 | 2.336 | 2.384 | 2.434 | 2.385 | 2.516 | 2.496 | 2.527 |  |  |
|                                                                        | Ebn at Noon |        | 960   | 966   | 968   | 926   | 885   | 880   | 882   | 898   | 885   | 918   | 918   | 935   |  |  |
|                                                                        | Edh at Noon |        | 89    | 102   | 111   | 129   | 137   | 128   | 122   | 115   | 118   | 98    | 92    | 87    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 3.81  | 4.78  | 5.80  | 6.60  | 6.72  | 6.96  | 6.62  | 6.49  | 5.33  | 4.83  | 4.07  | 3.58  |  |  |
|                                                                        | RadStd      |        | 0.38  | 0.51  | 0.44  | 0.44  | 0.30  | 0.43  | 0.53  | 0.35  | 0.51  | 0.43  | 0.41  | 0.31  |  |  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | -0.99 | -0.56 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page